Food Analysis

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food Analysis (A01766)

Short Title: Food Analysis
Faculty Science and Computing
Department Science
Cluster Food Science
Author EDUGGAN
Credits 5 **Level:** Introductory

Description of Module / Aims

Food analysis is an essential part of the agri-food industry, and encompasses the determination of nutritional value, compliance with labelling/legal requirements, assessment of product quality and detection of adulteration. The aim of this module is to introduce the student to the basic principles of analytical techniques commonly used to determine the chemical composition of foods. The module will provide the student with both theoretical and practical knowledge of techniques for the determination of macro- and micronutrients within foods. This module will equip students to work in food analysis across a broad range of agri-food industries in Ireland.

Programmes

FDSC-0001	BA (Hons) in Culinary Arts (WD_HCULA_B)	2 / 3 / M
FDSC-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	2 / 3 / M
FDSC-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 3 / M
FDSC-0001	BSc in Food Science with Business (WD_SFDSC_D)	2 / 3 / M

Indicative Content

- Quantitative analysis of foods for moisture, lipids, proteins, carbohydrates, vitamins and minerals
- Theory and practise of traditional wet chemistry techniques for food analysis
- Principles and application of spectrophotometric methods, including UV-visible, fluorescence and atomic absorption
- Principles and application of chromatographic methods in analysis, e.g. HPLC, GC

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the compositional components of a variety of food products.
2. Discuss the principles of and procedures for standard methodologies in food analysis.
3. Compare different analytical methods used within the food industry.
4. Employ a range of food analytical techniques for a variety of food types.
5. Report on a range of food analytical techniques for a variety of food types.

Learning and Teaching Methods

- Lectures will cover the principles of analytical techniques used in the food industry.
- Practical classes will enable students to perform food analysis experiments, analyse results and prepare reports.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	
Practical	50%	4,5

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of the learning outcomes. Lack of competence with regards to use of associated laboratory skills, techniques and instrumentation.

40%-49%: Demonstrate understanding of the fundamentals of the learning outcomes. Will display some ability with regard to the use of associated laboratory skills, techniques and instrumentation.

50%-59%: Demonstrate satisfactory understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to use of associated laboratory techniques instrumentation and good laboratory practice.

60%-69%: Demonstrate sound knowledge of the all topics contained in the learning outcomes and display good competence with regards to use of associated laboratory instrumentation and very good laboratory practice.

70%-100%: Demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to use of associated laboratory techniques and instrumentation and excellent laboratory practice.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Essential Material

- Nielsen, S.S. _Food Analysis_. 4th ed.. New York: Springer, 2010.

Supplementary Material

- Damodaran, S., K.L. Parkin and O.R. Fennema. _Fennema's__Food Chemistry_. 4th ed.. Boca Raton: CRC Press, 2007.
- Egan, H., R.S. Kirk and R. Sawyer. _Pearson's Chemical Analysis of Foods_. 8th ed.. Harlow: Longman Scientific & Technical, 1987.
- Pomeranz, Y. and C.E. Meloan. _Food Analysis: Theory and Practice_. 3rd ed.. London: Chapman & Hall, 1994.

Requested Resources

- Room Type: Chemistry Lab